

Programming and Data Structures Laboratory (CS19003)

Assignment 2 (Conditionals): Section 6

Department of CSE, IIT Kharagpur

21st January, 2026

General instructions to be followed strictly:

- Name your file as `<roll_no>_<assignment_no>_<question_no>`. For example, if your roll number is 14CS10001 and you are submitting assignment 3 question number (a), then name your file as 14CS10001_3-a.c
- Write your name, roll number, and assignment number at the beginning of your program.
- Use proper indentation in your code and comment. Marks may be deducted without proper indentation and comments.
- Submitting the **correct file** is your responsibility, any request of re-submission after the lab will **never** be entertained.
- **Zero tolerance** policy will be strictly followed for **plagiarism**. The marks will be put straight zero if such cases are identified.

Question-1 (10 points)

In a course, a student's result is determined using the following rules. A student is said to *pass* if at least one of the following conditions is satisfied:

1. The average of the two test scores is at least 50, or
2. The student has scored at least 60 in *both* tests.

However, there is an additional *exception rule*:

If the student has scored less than 30 in any one test, the student fails, regardless of the above conditions.

You are given two integers m_1 and m_2 , representing the marks obtained in two tests (each out of 100). Write a C program that determines whether the student passes or not. The solution must use only arithmetic, relational, and logical operators. **No loops, arrays, or pointers are allowed.**

Example 1:

(Input)

Enter the two test marks: 40 60

(Output)

The student passes

Example 2:

(Input)

Enter the two test marks: 25 80

(Output)

The student fails

Question-2 (20 points)

Electricity distribution companies calculate monthly electricity bills using different tariff categories depending on the type of consumer. Each consumer is assigned a tariff code, and the billing rules vary across these categories. Correct implementation of these rules is essential to ensure accurate and fair billing.

You are given two inputs: the number of electricity units consumed in a month (**units**) and an integer **tariff code** representing the consumer category. Your task is to compute the total electricity bill based on the tariff code using the rules described below.

- For **domestic consumers** (tariff code 1), the first 100 units are charged at Rs. 3 per unit and any remaining units are charged at Rs. 5 per unit.
- For **commercial consumers** (tariff code 2), all units are charged at a flat rate of Rs. 7 per unit.
- For **agricultural consumers** (tariff code 3), the first 200 units are free and any additional units are charged at Rs. 2 per unit.
- For **industrial consumers** (tariff code 4), the first 500 units are charged at Rs. 6 per unit and any remaining units are charged at Rs. 8 per unit.

Write a C program that reads the values of **units** and **tariff**, calculates the electricity bill according to the above rules, and prints the final bill amount as an integer. If an invalid tariff code is entered, the program should print -1. **No loops, arrays, or pointers are allowed.**

Example 1:

(Input)

Enter the units: 120

Enter the tariff code: 1

(Output)

Bill amount: 400

Example 2:

(Input)

Enter the units: 150

Enter the tariff code: 5

(Output)

Bill amount: -1

Question-3 (20 points)

Write a C program that determines whether a given calendar date is valid. The program should read three integers representing the *day* (d), *month* (m), and *year* (y).

A year is a *leap year* if it satisfies one of the following conditions:

$$(y \bmod 400 = 0) \quad \text{or} \quad (y \bmod 4 = 0 \text{ and } y \bmod 100 \neq 0)$$

In a leap year, the month of February has 29 days; otherwise, February has 28 days. The remaining months follow the standard calendar rules: April, June, September, and November have 30 days, while all other months have 31 days.

Your program must output 1 if the given date is valid, and 0 otherwise. The solution must use only arithmetic, relational, and logical operators. **No loops, arrays, or pointers are allowed.**

Example 1:

(Input)

Enter the date: 29

Enter the month: 2

Enter the year: 2024

(Output)

The date is valid

Example 2:

(Input)

Enter the date: 31

Enter the month: 4

Enter the year: 2022

(Output)

The date is not valid